Quark Flavor Equilibration of the Quark-Gluon Plasma

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Introduction: The Chemical Equilibration Gap

- The initial impact of a heavy-ion collision creates a highly energetic medium far from thermodynamic equilibrium.
- Gluons thermalize very rapidly (≈ 1 fm$/c$), but quarks and antiquarks take much longer (≥ 3 fm$/c$) to achieve chemical equilibrium.
- This delay leaves the early hydrodynamic QGP composition unknown.
- Article introduces a new viscous hydrodynamic model operating in *partial chemical equilibrium*.
- The medium is initialized as completely gluon-dominated, and local quark production is simulated over time.

Modeling Partial Chemical Equilibrium: Overview

Starting from a **pure-glue** QGP at τ_0, the model introduces a **time-dependent quark fugacity** (γ_q) during the hydrodynamic evolution. This is controlled by a single parameter: the **chemical equilibration time** (τ_{eq}).

What is varied?

- τ_{eq}: dictates how fast quarks are produced.
- The simulations compare $\tau_{\mathrm{eq}} = 0$ to 10 fm$/c$.

What is measured?

- Entropy production.
- Hadron yields and ratios.
- Flow observables: $\langle p_T \rangle$, v_2.
- Thermal photons

Main claim: The development of flow is sensitive to τ_{eq}, even when particle multiplicities are not.

Model (I): Hydrodynamics with a Chemistry-Dependent EoS

Standard hydrodynamics

$$\partial_\mu T^{\mu\nu} = 0$$

$$T^{\mu\nu} = \varepsilon u^\mu u^\nu - (P + \Pi)(g^{\mu\nu} - u^\mu u^\nu) + \pi^{\mu\nu}$$

- Fixed EoS in usual hydro: $P = P(\varepsilon)$

The modification

$$P = P(\varepsilon, \gamma_q)$$

- Same equations, but **chemistry-dependent EoS**

Quark fugacity γ_q

- $\gamma_q = 0$: pure glue (quark-poor)
- $\gamma_q = 1$: chemically equilibrated QGP
- $0 < \gamma_q < 1$: fewer degrees of freedom $\Rightarrow$ lower pressure

Physical consequence

Pressure gradients drive flow $\Rightarrow$ changing $P(\varepsilon)$ changes $\langle p_T \rangle$ and v_2.

Model (II): Nonequilibrium Equation of State

Construction

The EoS interpolates between two lattice results:

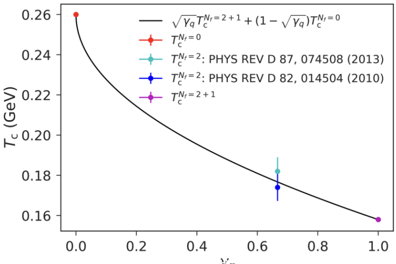
- **Pure glue** $SU(3)$ $(N_f = 0)$
- **Full QCD** $(N_f = 2 + 1)$

using the local quark fugacity $\gamma_q \in [0, 1]$.

Avoiding Two Transitions

Pure glue transitions at $T_0 \approx 260$ MeV, while full QCD crossovers at $T_3 \approx 158$ MeV. To prevent the fluid from experiencing two separate transitions, temperatures are rescaled using:

$$T_c(\gamma_q) = \sqrt{\gamma_q}\, T_3 + (1 - \sqrt{\gamma_q})\, T_0$$



Model (III): Quark Fugacity Evolution and τ_{eq}

1. The Equilibration Ansatz
Quark production is controlled by a simple function of local proper time (τ_p):

$$\gamma_q(\tau_p) = 1 - \exp\left(\frac{\tau_0 - \tau_p}{\tau_{\text{eq}}}\right)$$

2. Local Proper Time & Relativity
τ_p is advected with the fluid flow:

$$u^\mu \partial_\mu \tau_p = 1, \qquad \tau_p(\tau_0) = \tau_0$$

Because the expanding QGP droplet develops transverse flow, fast-moving cells experience **time dilation**.

3. The Takeaway
The periphery of the QGP system accumulates τ_p more slowly. Therefore, the edges chemically equilibrate *later* than the center in the lab frame.

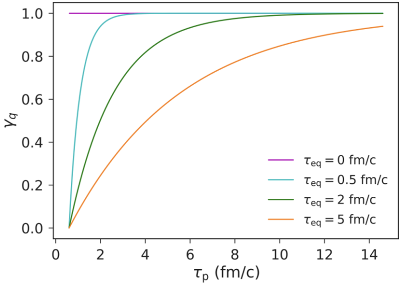


Fig. 3: $\gamma_q(\tau_p)$ saturating over proper time for different τ_{eq} values (with $\tau_0 = 0.6$ fm/c).

Hydrodynamic Evolution: Entropy Production

1. The Standard Assumption
In ideal hydrodynamics with a fixed Equation of State, entropy is strictly conserved.

2. The Nonequilibrium Effect
Because the medium is initialized as a pure-glue state, it begins with fewer effective degrees of freedom (lower initial entropy).

- As quarks chemically equilibrate, the degrees of freedom increase.
- This changing composition dynamically **generates entropy** during the fluid evolution.

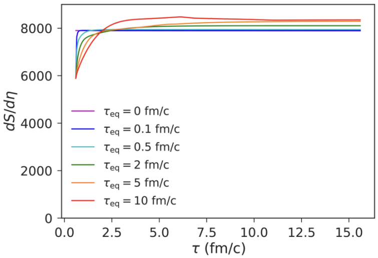


Fig. 5: Total entropy per unit space-time rapidity vs. time. Notice the initial deficit and subsequent growth for $\tau_{\rm eq} > 0$.

For larger $\tau_{\rm eq}$, the system starts with lower entropy and can produce up to $\approx 1/3$ of its final entropy dynamically.

How Far From Equilibrium at Particlization?

1. The Core Question

When the fluid cools and hadronizes (particlizes), is it already in chemical equilibrium ($\gamma_q \simeq 1$), or is it still quark-poor?

2. The Mechanism (Time Dilation)

Because γ_q grows with local proper time τ_p, the kinematics of the expansion matter.

- Fast-moving peripheral cells accumulate τ_p more slowly.
- Therefore, the outer regions equilibrate *later* than the core.

3. The Consequence

For large τ_{eq}, the system expands and cools faster than quarks can fully populate the edges. A significant fraction of the medium particlizes far from equilibrium ($\gamma_q \ll 1$).

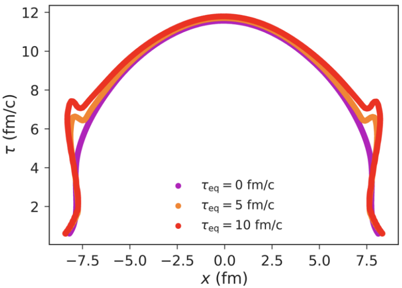


Fig. 9: The particlization hypersurface. Notice how large τ_{eq} forces the system to hadronize before quarks can fully equilibrate.

Hadron Production

1. The Observable

They measure the charged-particle multiplicity at midrapidity ($|\eta| < 0.5$), binned by centrality (N_{part}).

2. The Main Result

Varying the equilibration time τ_{eq} from 0 to 10 fm/c changes $dN_{\mathrm{ch}}/d\eta$ by only a few percent.

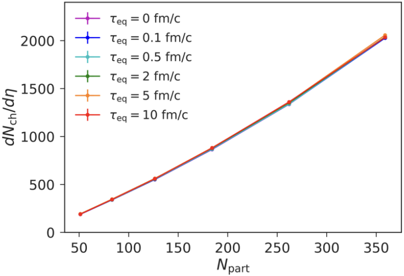


Fig. 12: $dN_{\mathrm{ch}}/d\eta$ vs N_{part}. The curves for different τ_{eq} nearly overlap due to the cancellation effect.

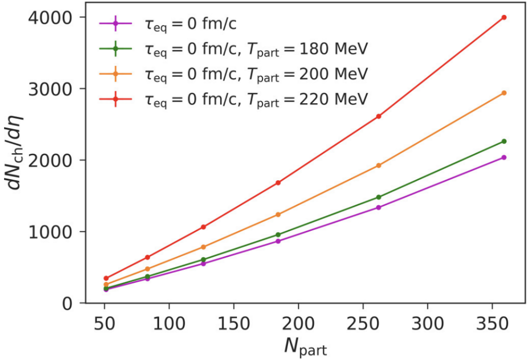
3. The Cancellation Mechanism

Why is it so insensitive? There are two strongly competing effects at the particlization hypersurface:

- **Suppression:** Larger τ_{eq} results in a lower γ_q, which naturally suppresses hadron yields.
- **Enhancement:** Larger τ_{eq} forces particlization at a higher critical temperature $T_c(\gamma_q)$, heavily enhancing thermal yields.

These two macroscopic effects essentially cancel each other out in the final state.

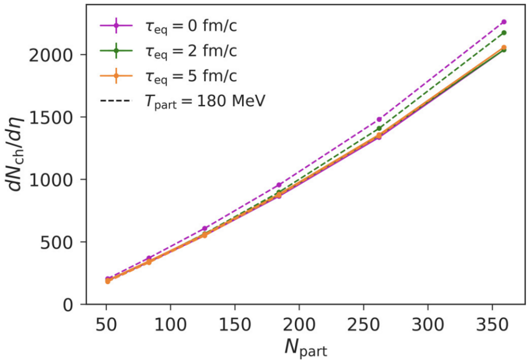
Charged-Particle Multiplicity: Varying T_{part} (0–60%)



Charged particle multiplicity for 0–60% centrality events evolved with varying particlization temperature T_{part} (each point corresponds to a 10% centrality bin).

- Here $\tau_{\mathrm{eq}} = 0$ fm$/c$ and T_{part} is varied.
- Increasing T_{part} increases $dN_{\mathrm{ch}}/d\eta$ at fixed N_{part}.
- The change is sizeable: higher T_{part} corresponds to an earlier/hotter particlization hypersurface, which boosts thermal yields.

Hadron Production: Multiplicity vs $N_{\rm part}$ at Fixed $T_{\rm part}$



Charged particle multiplicity for 0–60% centrality events evolved with varying $\tau_{\rm eq}$ at a fixed $T_{\rm part}$ (each point corresponds to a 10% centrality bin).

- Here $T_{\rm part} = 180$ MeV is fixed and $\tau_{\rm eq}$ is varied.
- The curves for different $\tau_{\rm eq}$ stay close: $dN_{\rm ch}/d\eta$ changes only by a few percent.
- This supports the cancellation picture: quark undersaturation suppresses yields, while the associated change in particlization conditions partially compensates it.

Identified Hadrons: π^+ and p Multiplicities vs N_{part}

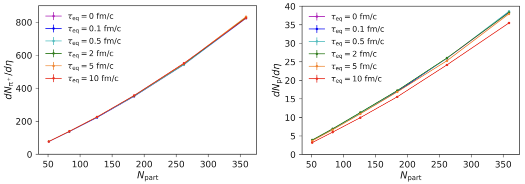
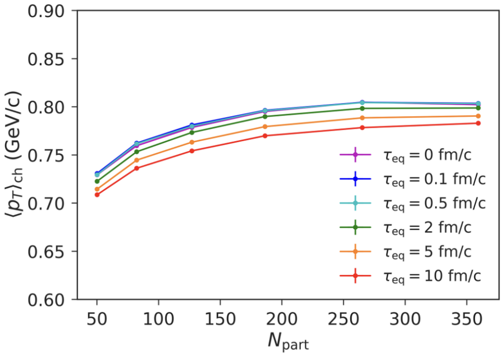


FIG. 14. Pion (left) and proton (right) multiplicities for 0–60% centrality events evolved with varying τ_{eq}. Each point corresponds to a 10% centrality bin.

- Left: $dN_{\pi^+}/d\eta$ vs N_{part} for different τ_{eq}.
- Right: $dN_p/d\eta$ vs N_{part} for different τ_{eq}.
- The π^+ curves nearly overlap across τ_{eq}.
- The proton yield shows a slightly larger sensitivity, with the largest τ_{eq} giving the lowest yields.

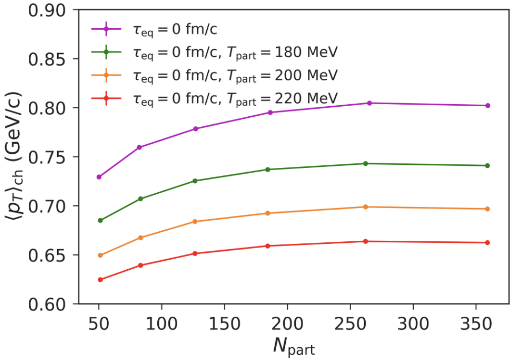
Radial Flow: $\langle p_T \rangle_{\mathrm{ch}}$ vs N_{part}



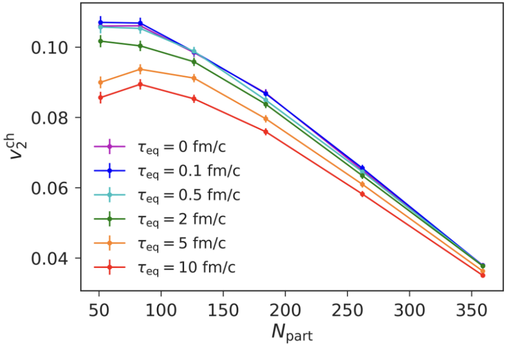
Each point corresponds to a 10% centrality bin within 0–60%.

- Observable: charged-hadron mean transverse momentum $\langle p_T \rangle_{\mathrm{ch}}$ as a function of N_{part}.
- Trend: $\langle p_T \rangle_{\mathrm{ch}}$ increases with N_{part} (stronger collective flow in more central events).
- Larger τ_{eq} systematically lowers $\langle p_T \rangle_{\mathrm{ch}}$.
- Slower chemical equilibration modifies the EoS.

Radial Flow: $\langle p_T \rangle_{\rm ch}$ and $T_{\rm part}$



- Here $\tau_{\rm eq} = 0$ fm$/c$ and $T_{\rm part}$ is varied.
- Increasing $T_{\rm part}$ lowers $\langle p_T \rangle_{\rm ch}$ at fixed $N_{\rm part}$.
- The ordering is monotonic: $T_{\rm part} = 180$ MeV gives the largest $\langle p_T \rangle_{\rm ch}$, while $T_{\rm part} = 220$ MeV gives the smallest.
- Earlier/hotter particlization leaves less time for transverse acceleration, reducing final radial flow.



- Observable: charged-hadron elliptic flow v_2^{ch} as a function of N_{part}.
- Centrality trend: v_2^{ch} decreases toward central collisions (large N_{part}).
- Sensitivity: larger τ_{eq} systematically reduces v_2^{ch} across centrality.
- Interpretation: delayed chemical equilibration modifies the pressure history and weakens the buildup of anisotropic flow.

Conclusion

- **Setup:** Hydrodynamics with a chemistry-dependent EoS $P(\varepsilon, \gamma_q)$.
- **Mechanism:** local proper time + time dilation $\Rightarrow$ outer regions equilibrate later.
- **Key result:** $dN_{\mathrm{ch}}/d\eta$ shows **cancellation** (yield suppression vs higher T_c).
- **Sensitivity:** $\langle p_T \rangle$ and v_2 show clearer dependence on τ_{eq}.

Future outlook

Flavor & system size

- Separate fugacities: **light vs strange** ($\gamma_{u,d}$, γ_s).
- Test in **smaller systems** (O+O peripheral A+A): shorter lifetime $\Rightarrow$ stronger sensitivity.

Constraints

- Add **fugacity-dependent viscosities** $\eta/s(\gamma_q)$, $\zeta/s(\gamma_q)$.
- Goal: **Bayesian calibration** to constrain τ_{eq} with other QGP parameters.